GAGE BALDASSARO

Portfolio | GitHub | LinkedIn | gb68@rice.edu | (504) 494-7226

Computer Science student focused on real-time rendering and gameplay systems

EDUCATION

Bachelor of Science in Computer Science | GPA: 3.83/4.0 August 2024 - May 2028
Rice University, Houston, TX
Courses Real-Time Rendering Systems, Algorithms, Computer Organization, Program Design, Machine Learning, Probability and Statistics, Linear Algebra, Multivariable Calculus, Differential Equations, Physics

SKILLS

Languages C++, C#, Java, Python, C, GLSL, HLSL, Lua, MATLAB
Engines and Tools Unity, Godot, Git, GitHub, Shader Graph, SteamWorks, APIs, OpenGL, Valgrind, JUnit

PROJECTS

OpenGL Renderer | Solo Project July - August 2026
C++, GLSL, OpenGL

- Engineered a custom multi-pass rendering pipeline from scratch using C++, OpenGL, and GLSL.
- Built a complex Blinn-Phong lighting and material system using tangent-space matrices for accurate normal mapping.
- Developed configurable post-processing shaders and robust 3D model loading via Assimp to render detailed environments and objects.

Code Athens **| Lead Programmer** July 2026
C#, Unity | Top 1% of 10.5k Submissions to GMTK Game Jam 2026 (Narrative, Audio)

- Led programming on a 6 person team of designers and artists to prototype, build, polish, and ship a complete visual novel game with interactive minigames within 96 hours.
- Rapidly prototyped and developed 6 drag-and-drop minigames and refined based on team feedback and playtesting.
- Designed an event-driven dialogue engine and a modular, asynchronous loading pipeline for seamless level transitions.

Mech Combat Demo | Solo Project May - July 2026
C#, Unity

- Developed a 3rd-person mech combat prototype inspired by *Armored Core VI*, owning character controls, camera, weapons, and enemy targeting end to end.
- Implemented a finite-state-machine based player movement controller for energy-limited dashing and hovering, fine-tuned for responsive, agile movement.
- Designed a data-driven weapon system with ScriptableObjects and quadratic-intercept lead targeting to hit moving enemies.
- Built a custom lock-on camera system with occlusion-aware target searching and velocity-based tilt.

EXPERIENCE

Unity Developer Intern | *Planetary Parfait* October 2025 - August 2026
Experiential Pixels Lab at Rice ECE, Houston, TX

- Shipped 20+ bug fixes and gameplay features to the public Steam build based on community feedback and team playtesting.
- Investigated and debugged key features of a large-scale Unity C# codebase through deliberate edge-case testing to ensure data validity.
- Engineered procedural terrain integration, collaborating with cross-functional teams to seamlessly render 2D Lunar data parsed from *JMARS* JSON API onto 3D Geospatial Lunar data from *Cesium*.
- Coauthored scientific publication (forthcoming) on immersive visualization in geospatial scientific workflows.

Electrical Engineering Intern June - July 2025
Petrotech Inc., New Orleans, LA

- Designed and documented a Modbus TCP/IP communication protocol enabling reliable PLC-to-HMI data exchange via DIGI One IAP.
- Programmed an Excel VBA HMI to monitor the PLC of an ALLISON 501-KB gas turbine to customer specifications, and built FactoryTalk View HMI screens from complex engineering drawings.